

Phase One Submission: Mercy Corps AgriFin Track

1. Team Name & Members

- **Team Name:** Technitians
- **Members:** Blessings Wanjiku (*Full-Stack Developer, AI Engineer*), Eugene Mutembei (*Data Scientist, Full-Stack Developer*)

2. Problem Route & Brief

- **Route:** Official Kenya AI Challenge AgriFin Brief
- **Selected Brief:** AFRACA – *The Invisible Farmer: Alternative Credit Scoring via Network-Based Trust & Climate-Resilient Risk Analysis.*

3. Problem Understanding & Bottlenecks

- **The Credit Gap:** Fewer than 20% of Sub-Saharan Africa's smallholders access formal credit because traditional FSPs treat unbanked farmers as financial ghost profiles due to a lack of formal bank statements, asset collateral, or conventional credit scores.
- **The Static Blindspot:** Current underwriting treats farmers as isolated, static entities. It completely ignores deep, informal socio-economic network capital, value-chain behaviors, and community peer-accountability systems.
- **Climate Volatility:** Traditional credit models fail to dynamically capture micro-climate volatility (e.g., localized droughts or flash floods), exposing lenders to unexpected portfolio defaults and trapping farmers in under-capitalization cycles.
- **RDBMS Architectural Bottleneck:** Aggregating relational data streams and shifting weather risks creates deeply nested SQL joins that break traditional relational database management systems (RDBMS). This causes crippling query latencies that make real-time alternative credit scoring operationally impossible.

4. Target User & Pain Point

- **Target User:** Credit Risk Officers, Underwriters, and Sacco/Microfinance Loan Managers at AFRACA-affiliated institutions.
- **The Exact Bottleneck:** Information asymmetry forces loan officers to manually reject thin-file applicants or rely on subjective guesswork. This results in slow turnaround times (2–4 weeks), high non-performing loan (NPL) rates, and an inability to scale portfolios safely.

5. Evidence & Research

- **The Collateral Disconnect:** Over 80% of East/West African smallholders utilize informal, relational networks (Chamas, cooperatives, mobile money), yet traditional credit rating bureaus capture 0% of this ecosystem data.
- **Predictive Power of Cooperatives:** Research from the *GIZ Data Governance in Africa Initiative* shows that structured cooperative crop-delivery logs and group accountability lower historical default rates below 5%.
- **Climate Default Drivers:** IPCC and regional meteorological data show unexpected agrometeorological shocks cause up to 75% of smallholder loan defaults across Sub-Saharan Africa.
- **Utility of Micro-Data:** The *World Bank's Digital Agricultural Payments Note* proves that linking micro-transaction velocities with localized climate maps provides an accurate proxy for predicting seasonal cash flows.

6. Proposed Solution

We are building **KilimoGraph Ledger**, an end-to-end, graph-augmented alternative credit-scoring platform linking a full-stack data engine with a point-and-click underwriting web interface.

- **Network Verification:** Automatically polls transactional streams (mobile money velocity, cooperative deliveries, agrovet purchase histories, and peer endorsements).
- **Micro-Climate Overlay:** Links farmer plot coordinates to historical and predictive climate indices to automate an environmental vulnerability index.
- **Value Proposition:** Replaces physical collateral with **Network Capital Collateral**, slashing loan approval cycles from weeks to under 5 minutes while safely mitigating risk.

7. AI, Data, and Neo4j Role

We model the agricultural economy as an interconnected, multi-dimensional knowledge graph in Neo4j to link financial data directly to environmental and social risk networks.

Advanced AI & Graph Architecture Pipeline

1. **Graph Data Science (GDS) Analytics:** Executes **Community Detection (Leiden/Louvain)** over cooperatives and **PageRank / Centrality** algorithms over `[VOUCHES_FOR]` relationship edges. Social capital from high-repayment community vectors programmatically propagates down the graph structure to calculate alternative credit ratings.
2. **Deterministic Cypher Extraction:** A backend multi-hop Cypher query extracts exact transaction velocities, cooperative compliance, peer weights, and local climate scores.
3. **Hallucination-Free GraphRAG:** Extracted graph paths feed directly into the context window of a fine-tuned LLM agent. The graph acts as the single source of truth, and the LLM translates native Neo4j relationships into plain-language underwriting justifications on the dashboard, completely eliminating hallucinations.

8. Build Plan

- **What We Will Build First (Days 1–2):** Deploy a cloud-hosted Neo4j Aura database instance and a data ingestion engine. We will write Python scripts using the official driver to seed 100 synthetic smallholder profiles, establishing our structural graph source of truth before building any visual or AI layers.
- **What the Prototype Will Demonstrate:** A live, end-to-end loan officer underwriting workflow: profile retrieval via unique Sacco/Phone ID; interactive node-graph visuals of community and wallet links; dynamic credit limits and climate risk dials calculated via graph algorithms; and an automated, plain-language GraphRAG justification detailing exact approval paths.
- **What Should Be Working (By June 27):** A cloud-hosted **Neo4j instance** executing multi-hop queries; a **FastAPI backend** handling alternative credit calculations; a **React/Tailwind web dashboard** rendering interactive visual widgets (Recharts); and an operational **GraphRAG pipeline** producing plain-language underwriting narratives.
- **What Will Be Mocked or Simulated:** Live telecom/mobile money transaction velocities (e.g., Safaricom M-Pesa webhooks) and real-time geospatial satellite data fetching (e.g., Google Earth Engine) will be simulated using pre-computed metadata fields embedded directly into graph nodes.

9. Key Assumptions, Risks, and Responsible AI Use

- **Core Assumption:** Smallholders are active in at least one community network that tracks activity (Saccos, agrovets, Chamas, or registered SIM cards).
- **Technical Risks:** Managing unstructured data formats from remote cooperatives, last-mile rural connectivity drops, and systemic correlated defaults from extreme, regional climate shocks.
- **Responsible AI & Bias Mitigation:** Enforces a strict **Human-in-the-Loop** model—the platform never issues automated rejections. Powered by deterministic GraphRAG backed by immutable Neo4j path histories, all scores are transparent and auditable, allowing officers to explain precisely how farmers can improve their rating.

10. Support Needed

- **Domain Expert Feedback:** Mentorship from AFRACA underwriters or Mercy Corps AgriFin experts to validate mathematical scoring weights balancing social network capital against weather parameters.
- **Neo4j Performance Tuning:** Technical guidance to optimize multi-hop Cypher queries and index performance over dense community node clusters.